Auteur: Hina Rana
Studierichting: Informatica
Oefeningenreeks 2B nr. 1.2

$\sin(Bgtanx)$ neem $y = \text{Bgtan}(x)$ met $y \in \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[$

$\tan(y) = x$

$\tan y \cdot \cos(y) = x \cdot \cos(y)$

$\sin y = x \cos(y)$

$\sin^2 y + \cos^2(y) = 1$

$(x \cos y)^2 + \cos^2(y) = 1$

$x^2 \cos^2 y + \cos^2(y) = 1$

$\cos^2 y(x^2 + 1) = 1$

$\cos^2 y = \frac{1}{x^2 + 1}$

$\cos y = \frac{1}{\sqrt{x^2 + 1}}$

$x \cos y = \frac{x}{\sqrt{x^2 + 1}}$

$\sin y = \frac{x}{\sqrt{x^2 + 1}}$

References